VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama"Belagavi, Karnataka-590018

PROJECT REPORT ON

"FORMULATION OF NOURISHING HAIR DYE USING *Areca catechu L.* PEDUNCLE EXTRACT FOR STRONG AND LUSTROUS HAIR"

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

BACHELOR OF ENGINEERING
In
BIOTECHNOLOGY

Submitted by

Ms. Anusha	(4GM21BT003)
Ms. Rinku K	(4GM21BT023)
Ms. Shreya R Kinnal	(4GM21BT030)
Ms. Summaiya Kouser	(4GM21BT037)

Under the guidance of

Dr. GURUMURTHY H
Professor
Department of Biotechnology

DEPARTMENT OF BIOTECHNOLOGY
G M INSTITUTE OF TECHNOLOGY, DAVANGERE-577006

(Affiliated to Visvesvaraya Technological University, Belagavi, Recognized by AICTE and Govt. of Karnataka and Accredited by NBA, New Delhi) 2024

G M INSTITUTE OF TECHNOLOGY,

DAVANGERE-577006

DEPARTMENT OF BIOTECHNOLOGY

CERTIFICATE

This is to certify the project work entitled **"FORMULATION OF NOURISHING HAIR DYE USING *Areca catechu L.* PEDUNCLE EXTRACT FOR STRONG AND LUSTROUS HAIR"** carried out by **Ms. Anusha (4GM21BT021)), Ms. Rinku K (4GM21BT023), Ms. Shreya R Kinnal (4GM21BT030), Ms. Summaiya Kouser (4GM21BT037)** a bonafide students of GMIT Davangere in partial fulfillment for the award of Bachelor of Engineering / Bachelor of Technology in Biotechnology of the Visvesvaraya Technological University, Belagavi during the year 2024-2025. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said Degree.

Signature of Guide	Signature of Co-ordinator	Signature of HOD	Signature of Principal
Dr. Gurumurthy H	Dr. Gurumurthy H	Dr. Prakash K K	Dr. Sanjay Pande M B
Professor	Professor	Professor and Head	GM Institute of Technology

Name of Examiners	Signature of Examiners
1	1
2	2

DECLARATION

We hereby declare that this project work entitled "**FORMULATION OF NOURISHING HAIR DYE USING *Areca catechu L.* PEDUNCLE EXTRACT FOR STRONG AND LUSTROUS HAIR**"is a group study conducted by our team, under the guidance of Dr. Gurumurthy H, Professor, Department of Biotechnology, GMIT, Davangere submitted in partial fulfillment of the requirements of the degree of B.E BIOTECHNOLOGY of Visvesvaraya Technological University, Belagavi during the year 2024-2025.

We further declare that this work has not formed the basis for the award of any degree / diploma of any other university/institution.

Ms. Anusha	(4GM21BT003)
Ms. Rinku K	(4GM21BT023)
Ms. Shreya R Kinnal	(4GM21BT030)
Ms. SummaiyaKouser	(4GM21BT037)

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ABSTRACT

Hair dye formulations typically use synthetic chemicals, which can lead to damage or unwanted reactions. This study aims to create a natural and nourishing hair dye using Areca catechu L. peduncle extract. This extract is abundant in bioactive compounds such as catechins and epigallocatechin, which are known for their strong 5-alpha reductase inhibitory activity, helping to prevent male pattern baldness. The resulting hair dye not only delivers effective color but also nourishes the scalp and strengthens the hair. Evaluations showed excellent color retention, enhanced hair strength, and a beautiful shine, positioning Areca catechu peduncle as a promising natural option for hair care and loss prevention.

Keywords: *Areca catechu, catechin, epigallocatechin, 5-alpha reductase inhibitor, male pattern baldness.*

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1. INTRODUCTION

The Areca palm (Areca catechu L.), often referred to as the areca nut palm, is a tropical tree that holds significant cultural, medicinal, and economic value, particularly in South and Southeast Asia. This palm tree yields the areca nut, also known as betel nut, which has been utilized for over 10,000 years. While the nut is most commonly consumed as a chewable stimulant, frequently combined with betel leaves and lime, the areca plant has diverse applications, especially in traditional medicine and various industrial fields. Despite its historical importance and ongoing use, the areca nut is currently facing scrutiny due to health concerns, particularly its link to oral cancers. Nevertheless, the areca nut continues to be a vital agricultural product and a crucial part of rural economies, especially in India.

Geographical Distribution in India

Areca catechu is primarily cultivated in tropical and subtropical areas. In India, the production of areca nuts is most concentrated in the states of Karnataka, Kerala, Assam, Tamil Nadu, and West Bengal. Karnataka, particularly in the districts of Dakshina Kannada, Uttara Kannada, and parts of Sirsi, is recognized as the leading producer of areca nuts in the nation. The plant flourishes in regions characterized by high humidity, well-drained soil, and temperatures between 25°C and 35°C, making the coastal areas of South India particularly suitable for its growth. Areca nuts are cultivated both in dedicated plantations and as part of mixed cropping systems alongside other agricultural products such as coconut, banana, and pepper.

Botanical Classification

The botanical classification of Areca catechu is as follows:

Kingdom: Plante
Division: Angiosperms Class:Monocots
Order: Arecales
Family: Arecaceae (Palm family)
Genus: Areca
Species: A. catechu

Areca catechu is a medium-sized, evergreen tree that can grow up to 20 meters tall. Its trunk is slender and smooth, and the palm features large, feathery leaves. The fruit, or areca nut, is oval in shape.

The main chemical components of areca nut are quite complex, with various constituents found in different parts of the nut (see Figure 1). The primary components include polyphenols (10–30%), polysaccharides (18–25%), fibers (10–15%), fatty acids (10–15%), and alkaloids (0.3–0.7%) . The husk of the areca nut is mainly made up of cellulose and has a higher lignin content, which contributes to its increased hardness and the mechanical friction it produces on the oral mucosa during chewing. This makes it a valuable resource for wood fiber. Additionally, areca nut contains a variety of bioactive substances. Alkaloids are nitrogen-containing organic compounds that are typically found in higher plants, including families such as Papaveraceae, Loganiaceae, Leguminosae, Ranunculaceae, and Menispermaceae. Many of these compounds are cyclic nitrogenous substances and natural psychoactive agents. The total alkaloid content in areca nut ranges from 0.3% to 0.7%. The four main alkaloids present in areca nut are arecoline (N-methyl-1,2,5,6-tetrahydropyridine-3-carboxylic acid methyl ester), arecaidine, guvacoline, and guvacine, with their concentrations in fresh seeds being 0.30–0.63%, 0.31–0.66%, 0.03–0.06%, and 0.19–0.72%, respectively. Arecoline is an amphoteric compound that belongs to the pyridine alkaloids and was first isolated from areca nut in 1888 by German pharmacist E. Jahns. It is often found in combination with tannins, which means it is typically extracted by soaking in ammonia before being extracted with ethanol, ether, or chloroform. Notably, arecoline has a structure similar to nicotine and exhibits strong stimulative effects. After alcohol, caffeine, and nicotine, arecoline ranks as the fourth most widely used addictive substance globally.

Tannins are a group of water-soluble polyphenolic compounds with intricate structures, commonly found in legume seeds, cereals, cacao, vegetables, coffee, tea, berries, and nuts. They are derivatives of flavanols and are primarily divided into hydrolyzable tannins and condensed tannins. In areca nuts, the predominant type is condensed tannins, which are more plentiful in unripe nuts and diminish as they ripen. Tannins typically have an astringent and bitter taste, contributing to the bitterness of areca nuts. The extraction of tannins is usually performed through solvent extraction, utilizing substances like water, ethanol, methanol, chloroform, and acetone. Research has refined the extraction process for tannins in areca nuts, recommending the use of 80% acetone, adjusting the pH to 4, and extracting for 90 minutes. Previous studies have identified various tannins extracted from areca nuts, including (+)-catechin, (−)-epicatechin, procyanidin B1, procyanidin B2, areca tannin A1, areca tannin B1, areca tannin B2, and areca

tannin C1.Both areca husk and peduncle are by-products of areca nut (betel nut) cultivation, and they contain tannins, which are polyphenolic compounds. These agricultural wastes have potential applications, particularly due to their tannin content and other bioactive compounds. Areca husk, the outer covering of the areca nut, is known to have significant amounts of tannins, along with components like cellulose, hemicellulose, lignin, and other polyphenols. The tannins found in areca husk are typically condensed tannins (proanthocyanidins), which possess antioxidant properties and can show antimicrobial activity. These tannins may help inhibit certain enzymes, such as 5-alpha reductase, much like other tannin-rich substances such as catechins. Similarly, the peduncle also contains polyphenols, including tannins that may offer antioxidant, antimicrobial, and other beneficial effects. The concentration of tannins can vary based on the maturity of the peduncle and the specific variety of areca nut.

Fig 1: Areca Peduncle

Fig 2: Areca Husk

Tannin compounds like catechins and epigallocatechin gallate (EGCG) have been found to inhibit 5-alpha reductase, the enzyme that converts testosterone into dihydrotestosterone (DHT). DHT is a powerful androgen linked to issues like hair loss (male pattern baldness) and benign prostatic hyperplasia. Catechins, especially those in green tea, along with EGCG, a key polyphenol in green tea, have been researched for their ability to inhibit 5-alpha reductase activity. By blocking this enzyme, these compounds may help lower DHT levels, potentially offering benefits for conditions related to high DHT, such as hair loss or prostate problems. While studies indicate that these polyphenols can serve as natural inhibitors of 5-alpha reductase, their clinical significance and effectiveness are still being explored.

2. OBJECTIVES

1. **To extract phytochemical components from Arecanut peduncle and Husk.**
2. **To perform Biochemical analysis for presence of Tannin.**
3. **To formulate and characterize the Hair Dye.**

3. LITERATURE REVIEW

1. Huihui.S , Wenzhen.Y et al, Bioactive Components of Areca Nut: An Overview of Their Positive Impacts Targeting Different Organs. *Nutrients*. 2024 Feb 29; 16(5), 695

This study explores the intricate chemical makeup of areca nut, which includes alkaloids, tannins, flavonoids, polyphenols, polysaccharides, and fatty acids. These bioactive compounds demonstrate a range of functions, such as antibacterial, deworming, antiviral, antioxidant, anti-inflammatory, and antitumor effects. Found in various parts of the nut, these phytochemicals contribute to the protection of different organs and physiological systems, showcasing their therapeutic potential. The research underscores the significance of areca nut as a valuable resource for developing pharmaceuticals, nutraceuticals, and functional foods aimed at combating infections, oxidative stress, inflammation, and chronic diseases.

2. Chougule .K,Chougule.M et al, Indian herbs that act as 5-alpha reductase inhibitors. *International journal of pharmaceutical, chemical and biological sciences,* 2017; 7(3), 265-

This article discusses several Indian herbs that may inhibit the activity of the 5-alpha-reductase enzyme, which plays a crucial role in androgenic alopecia. By targeting this enzyme, these herbs could potentially decrease the conversion of testosterone to dihydrotestosterone (DHT), a hormone associated with hair loss. The herbs identified contain phytochemicals that are beneficial for hair health, suggesting a natural alternative for addressing hair thinning and loss. This perspective not only highlights the therapeutic promise of traditional herbal remedies but also paves the way for further investigation into their effectiveness and mechanisms in promoting hair growth and overall scalp health.

3. Azizi .A, Mumin.N.H et al, Phytochemicals with Anti 5-alpha-reductase Activity: A Prospective for Prostate Cancer Treatment. *F1000Research* .2021; 5(10), 221

The conversion of testosterone into dihydrotestosterone (DHT) is facilitated by the enzyme 5-alpha-reductase. This enzyme has two isoforms, type 1 and type 2, which are present in various tissues such as the prostate, skin, and hair follicles. By converting testosterone, 5-alpha-reductase introduces a hydrogen molecule, resulting in DHT, a more powerful androgen. DHT is crucial for the development of male secondary sexual characteristics and prostate growth. However, high levels of DHT are associated with issues like androgenic alopecia (hair loss), benign

prostatic hyperplasia (BPH), and acne. To address these DHT-related conditions, inhibitors of 5-alpha-reductase are commonly used in treatment.

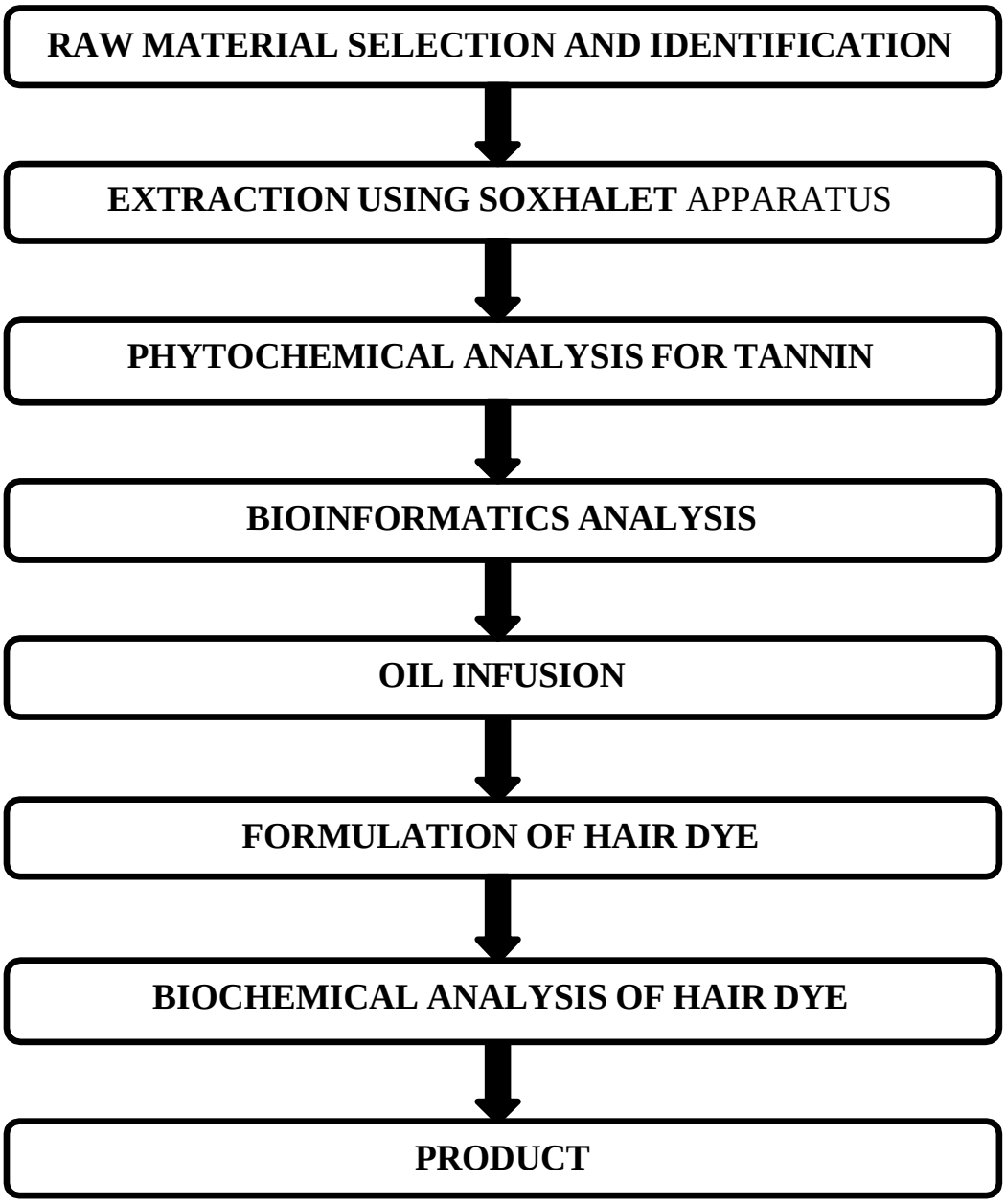
4. Ustuner E.T. Cause of androgenic alopecia: crux of the matter. *Plast Reconstr Surg Glob Open*. 2013; 1(7), 64.

This paper delves into the paradox of dihydrotestosterone (DHT) being linked to hair loss in androgenic alopecia (AGA) while simultaneously encouraging hair growth in other body areas. It details how DHT binds to androgen receptors in hair follicles, triggering a miniaturization process in specific genetically susceptible regions of the scalp, particularly on the crown. Over time, this leads to hair that is shorter, thinner, and less pigmented. The distinct balding pattern is attributed to variations in androgen receptor density, the activity of the 5-alpha-reductase enzyme, and genetic factors in these scalp areas. The paper highlights the selective effects of DHT, providing insight into the molecular mechanisms underlying AGA.

5. Guo.Z, Wang.Z et al, Extraction and identification of bioactive compounds from areca nut (Areca catechu L.) and potential for future applications, *Food Frontiers*. 2024; 1–24.

The process of extracting polyphenols from the husk of Areca catechu involves using heated reflux-assisted ether solvent extraction or soxhlet extraction with 20 g of crushed sample and 80% ethanol at 80°C. The extract is then analyzed through UV spectrophotometry, where 0.3 mL of a 1 mg/mL areca extract solution is combined with distilled water and ferrous tartrate at pH 7.5, and the absorbance is measured at 540 nm. Additionally, high-performance liquid chromatography (HPLC) is utilized, employing acetonitrile and 0.3% formic acid, with detection at 280 nm. The total polyphenol content varies from 9.8 to 58.8 µg/mL, with catechin levels ranging from 1.3% to 5.9% and epicatechin from 0.77%to2.4%.

4. METHODOLOGY



5. RESULT AND DISCUSSION

5.1 EXTRACTION PROCESS USING SOXHLET APPARATUS

Fig 3: Flowchart Depicting Phytochemical Extraction

COLLECTION:

Collect areca nut husk and peduncle from the freshly harvested or ripe areca nut trees. Make sure the materials are clean of any visible signs of decay, pests and contamination.

CLEANING:

Scrub the husk and peduncle under clean running water until surface dirt and debris has been removed. When washing, do not use detergent or chemicals they will interfere with the extraction.

DRYING:

Single layer spread of cleaned husk and peduncle on a clean surface, if possible on a drying tray or mesh to facilitate even drying.

Shade Drying:

To avoid photo degradation of bioactive compounds, dry the materials in a shaded, well-ventilated area.Depending on humidity levels, this can take 7–10 days.

Oven Drying (Optional):

The materials are placed in a hot-air oven and dried to absolute dryness at 50–60°C for 6–8 h. And check for dryness by breaking the material, which should snap easily when bent.

GRINDING:

Use a mechanical grinder or pulverize to pulverize the dried husk and peduncle to a fine powder. Pass through a 40-mesh sieve to obtain powder of uniform size — this is critical for maximum extraction.

WEIGHING THE SAMPLE

Weigh exactly 50 grams of the prepared powder using a calibrated electronic balance.

Well, this powder which can be used for anything from food to medicine — is precisely what you need, and you transfer it to a clean, dry receptacle for use.

SOXHLET APPARATUS

Solvent Addition:

Add 200-250 mL of 95% ethanol to the round bottom flask. It should be just enough to provide continuous extraction and not too much, to fill the flask all at once.

Sample Placement:

Blend 50 g of powdered arecanut husk and peduncle and pack it in cellulose or filter paper thimble. Do not try to pack the column too full so that the solvents can to through it effectively. Insert the thimble into the Soxhlet extractor making it compact without blocking the channel.

Assembling the Apparatus:

After that, fit the Soxhlet extractor to the round-bottom flask, and place the condenser on top of it. All the joints should be appropriately lubricated with grease to minimize on vapor leakage.

Safety Precautions:

This should be done while in a fume hood or any well-ventilated area since ethanol used ignited. Fix all the components with clamps to prevent any dangers through heat.

OPERATION OF THE SOXHLET EXTRACTION

Heating:

Put the round-bottom flask in a heating mantle or a water bath.

Heat it to ensure the reflux (best around 60-70 degrees centigrade or sub boiling point of ethanol of 78.5 degrees centigrade).

Extraction Cycles:

After distillation allow the ethanol to evaporate, condense in the condenser, and then collecting ethanol in the Soxhlet extractor. The solvent filters through the sample dissolving the extractable compounds in it. If the solvent covers the extraction chamber and returns into the flask siphoning, then a cycle is done. Cycles should be performed 15 times to obtain the highest yield of bioactive compounds.

Monitoring:

Moreover, sometimes check for any such sign of solvent loss or overheating.If the volume of the solvent decreases significantly during process, add the fresh solvent.

COLLECTION OF THE EXTRACT

Cooling:

After 15 cycles turn off the heating mantle and allow the entire apparatus to cool to room temperature before continuing.

Transfer Extract:

Proper demising of this apparatus is necessary. To another clean, labelled glass container, pour the ethanol extract using a funnel from the round-bottom flask. Note that solvent removal (concentration of extract) is a crucial step given the high polarity of substances used in this study.

Final drying:

Drying the extract using a water bath involves evaporating the solvent to obtain the concentrated or dried extract.

5.2 QUALITATIVE ANALYSIS

PROCEDURE:

Step 1: Preparation of the Ferric Chloride Solution

If not already prepared, dissolve 10 g of Ferric chloride in 100 mL of distilled water to create a 10% Ferric chloride solution. Stir thoroughly until the Ferric chloride is completely moistened or dissolved. The resulting solution should appear light yellow.

Step 2: Preparation of the Sample:

If the sample extract is not ready, create an aqueous solution by dissolving the extract in distilled water. If there are any insoluble particles, filter them out of the solution.

Step 3: Addition of Extract

Using a dropper or pipette, measure 2 mL of the sample extract and transfer it into a clean test tube.

Step 4: Addition of the Ferric Chloride Solution

Add 3 to 5 drops of the 10% Ferric chloride solution into the test tube containing the sample extract.

Step 5: Observation of Colour Change

Immediately observe any color change in the solution: a blackish-blue color indicates the presence of tannins.

Fig 4: Blackish blue colour observed indicating the presence of Tannin

5.3 QUANTITATIVE ANALYSIS

STANDARDISATION OF TANNIC ACID STOCK SOLUTION

Weighing the Tannic Acid:

Take 100 mg of tannic acid with the help of analytical balance and prepare a 100 mg/L standard solution. It is recommended to make sure that there is no slop for easy measurements.

Dissolving Tannic Acid:

Dilute the tannic acid to volume in a 100 mL volumetric flask. Finally, turn on the water at about 50mL and mix gently until all tannic acid lyses is in solution.

Dilution:

Finally, after dissolution, make up the volume up to 100 ml with distilled water. After that preferably blend until uniformity is achieved.

Labeling:

You need to label this flak with the title, Standard Tannic Acid Solution (100 mg/L) merely to make the labelling clear.

Fig 5: Standard Tannic Acid Preparation

HEAT REFLUX OF ARECANUT EXTRACT

Weighing the Extract:

Using 50 g of prepared arecanut husk and peduncle extract in the required quantity. Usually for initial refluxing, 10 g of the crude extract is utilised.

Reflux Setup:

Put the extract in a round bottom flask and then add to it 200ml of distilled water.Prepare a reflux apparatus where all joints shall be greased tightly so as to reduce vapor loss recurrent to the fractionation step.

Heating:

Place the mixture under heating mantle/ water bath and heat the mixture at 60-70°C for 1 hour.It is advisable to stir occasionally to avoid a situation where some parts of the dough are overcooked than others.

Cooling:

For 1 hour, leave the solution and then let it cool to a normal temperature in readiness for further processing.

Fig 6: Heat Reflux of Areca husk &pedunle Extract

DILUTION OF THE EXTRACT

Measuring the Extract:

With a pipette, transfer 5 mL of the cooled refluxed extract and pour the content into a petri dish.

Dilution:

Pour the extraction into a 100 mL volumetric flask through the receiver column using 5mL of extract.This should preferably be done to the mark with distilled water to ensure that tannins spread out evenly in solution.

Labelling:

Finally, place a label with a majuscule inscription "Diluted Extract Solution" for identification purposes upon the volumetric flask.

PREPARATION OF REAGENTS

Ferric Chloride Solution (1%):

Take 1g of ferric chloride inside the analytical balance and weigh it. Pour 100 mL of distilled water into a clean beaker and dissolve it in the water. Stir again until dissolved forms a solution with the other ingredients.

Ferricyanide Solution (1%):

0.5 g of potassium ferricyanide was accurately weighed with an analytical balance. Dissolve it in 100ml distilled water in a clean beaker .Stir until fully dissolved if using powder forms; when using liquids gently combine. Keep it in a reagent bottle which should be labelled.

Reaction Mixture Preparation:

Prepare 5 mL of the diluted extract solution and add 5 mL of it to whichever test tube or cuvette is available. Now, 1mL of 1% ferric chloride solution and 1 mL of 1% ferricyanide solution were added into the test tube.Mix the components until the colour forms and leave the mixture to work at room temperature for about 10–15 minutes.

Standard Reaction Mixture:

Perform the standard reaction preparation same like the previous steps but use 5 mL of standard tannic acid solution instead of the sample extract.

Blank Preparation:

It also recommends the use of distilled water as the blank solution for calibration.

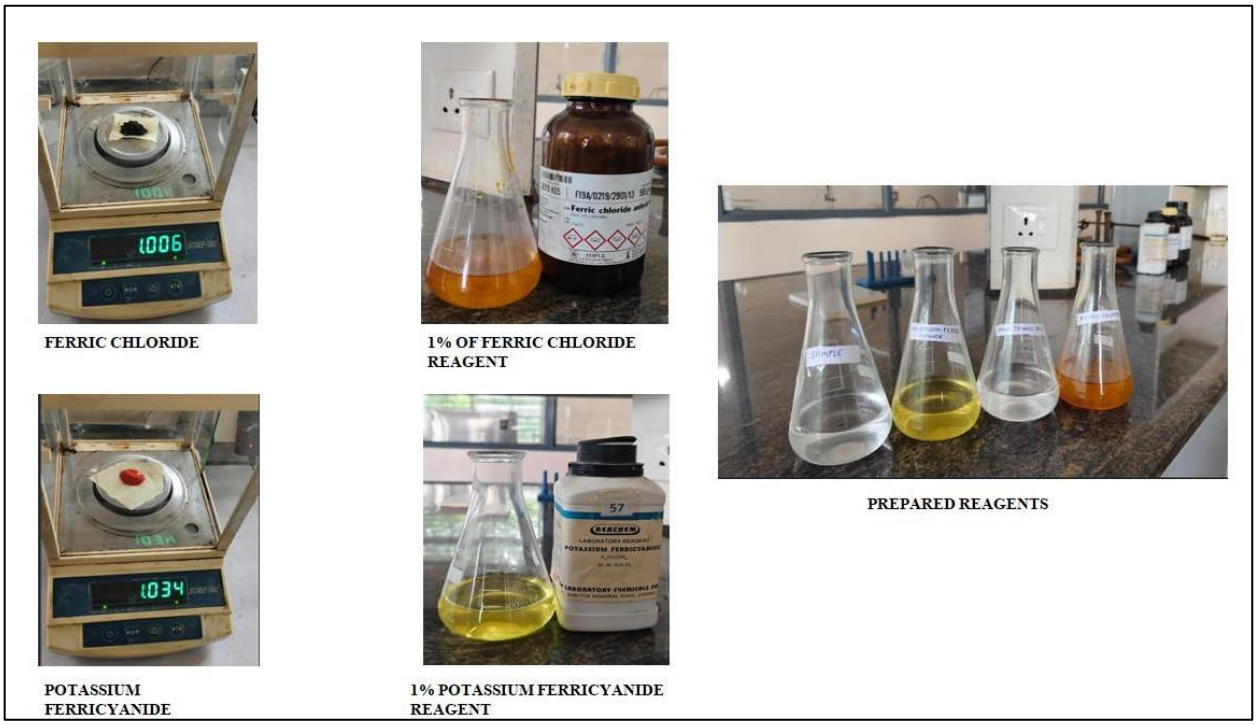


Fig 7: Preparation of Reagents

UV SPECTROSCOPY ANALYSIS

Instrument Setup:

The UV-Vis spectrophotometer should be switched on half an hour earlier to warm up before use.The absorbance conditions of the tannin-ferric complex were achieved by setting the wavelength at 720 nm, where the tannin complex with maximum absorbance to the ferric ion was expected.

Calibration with Blank:

Take a quartz cuvette, clean it and then pour a blank solution into the cuvette.This is in order to set the cuvette holder in the spectrophotometer and adjust the reading to zero.

Measuring Standard Absorbance:

Put another cuvette with the standard tannic acid reaction mixture.They determined the absorbance at 720 nm of the prepared sample. The standard absorbance should be 0.501 nm.

Measuring Sample Absorbance:

Transfer a third cuvette with distilled water and then add the sample reaction mixture.The absorbance should be measured at 720 nm. The obtainable sample absorbance should be 0.475 nm.

The percentage of Tannin content is been calculated below.

Fig 8: Sample Absorbance at 720nm in UV

Fig 9: Standard tannic acid Absorbance

FORMULA:

% OF TANNIN PRESENT=$\frac{SAMPLE\ ABSORBANCE}{STANDARD\ ABSORBANCE} X \frac{STANDARD\ CONCENTRATION}{SAMPLE\ CONCENTRATION} X\ PURITY\ OF\ STANDARD$

Standard Concentration After Dilution	**0.005**
Sample Concentration	**0.2**
Standard Absorbance	**0.501**
Sample Absorbance	**0.475**

Table 1: values of concentration and Aborbance

$$\%\ OF\ TANNIN\ PRESENT = \frac{0.475}{0.501} X \frac{0.005}{0.2} X98 = 2.32\%$$

5.4 BIOINFORMATICS ANALYSIS

TARGET SELECTION

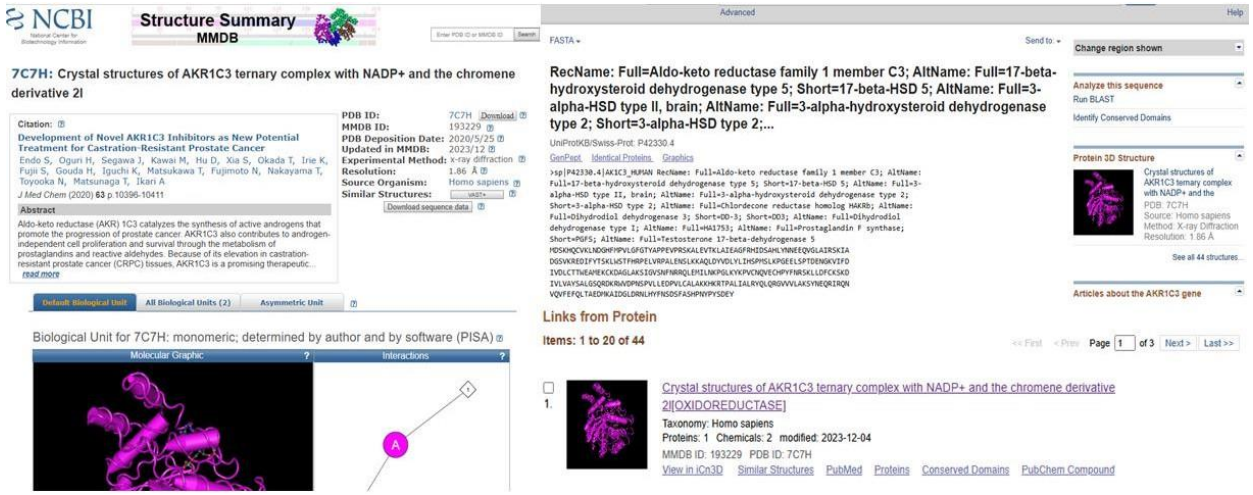


Fig 10: Alpha Reductase in Humans

The target used for this study is 7C7H which is a 5 alpha reductase found in human.

The 3D structure of 5-alpha-reductase was obtained from the Protein Data Bank (PDB ID: 7C7H).

LIGAND SELECTION

Selection of Ligands:

Catechin

Epicatechin Gallate

Epicatechin

The structures of these ligands were obtained from the PubChem database in .sdf or .mol format and then converted to .pdb format using Open Babel.

CATACHIN, EPICATACHIN AND EPICATACHIN GALLATE DOCKED WITH 5 ALPHA REDUCTASE AS TARGET.

Docking Simulation

Swiss Dock Platform:

The prepared 7C7H PDB file along with the ligands (catechin, Epicatechin gallate, and epicatechin) were uploaded to the Swiss Dock server. The docking parameters were set to standard precision to assess the binding affinity and interactions.

Output Analysis:

The binding modes and the best docking scores (ΔG, kcal/mol) were obtained.

Discussion

Binding Affinity and Structural Changes

Catechin, Epicatechingallate, and Epicatechin successfully docked into the active site of 5-alpha-reductase. The binding scores indicated a strong affinity, with epicatechingallate exhibiting the highest binding energy (most negative ΔG value). Key residues in the active site interacted with these ligands through hydrogen bonding and hydrophobic interactions, resulting in conformational changes in the protein structure.

Deactivation Mechanism

The docking simulations showed that the binding of tannin and its derivatives led to considerable structural changes in the active site of the enzyme. These alterations likely disrupted the normal catalytic activity of 5-alpha-reductase, effectively deactivating the enzyme. Structural distortion was noticeable in the alpha-helices and beta-sheets surrounding the active site, particularly in residues crucial for substrate recognition and binding.

Visualization of Structural Changes

The superimposed structures of the native enzyme and the enzyme-ligand complexes showed a clear shift in the active site's geometry.

This structural alteration supports the hypothesis that binding of these ligands inhibits the enzymatic activity by steric hindrance or allosteric effects

CATACHIN

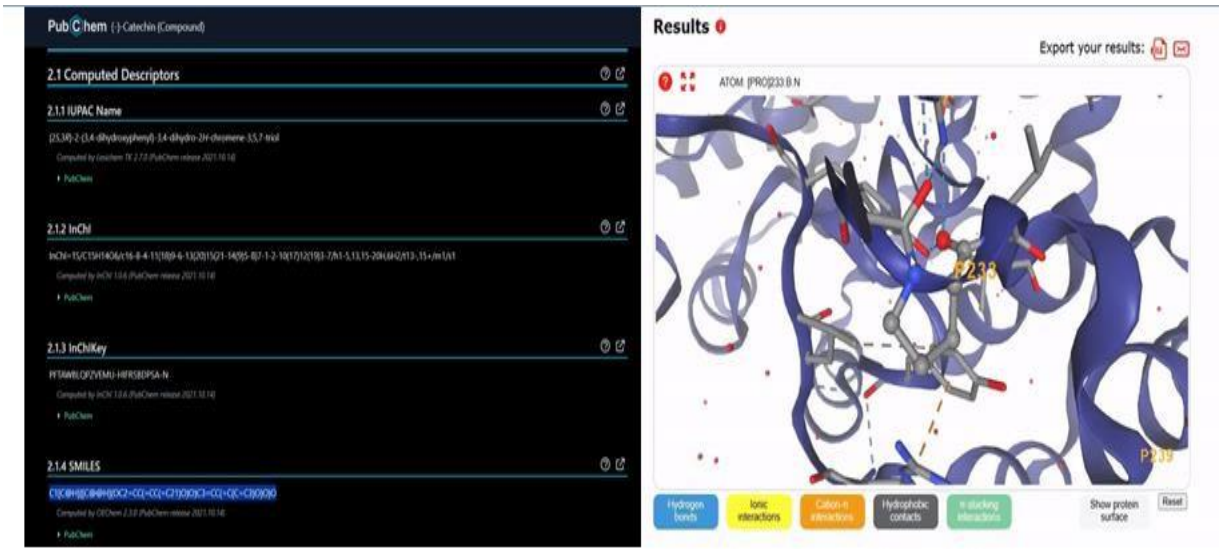


Fig 11: Catechin Smilies identified & docking with 5 Alpha Reductase

CALCULATED AFFINITY (Kcal/mol)

MODEL	AFFINITY
1	-6.27
2	-6.02

HYDROGEN BONDS	LYS-ASP
	GLU-LYS
	TYR-ASP
	ALA-ILE
	PHE-LYS
	HIS-TYR

Table 2: Calculated Affinity (Kcal/Mol) And Hydrogen Bonds Identified

EPICATACHIN GALLATE

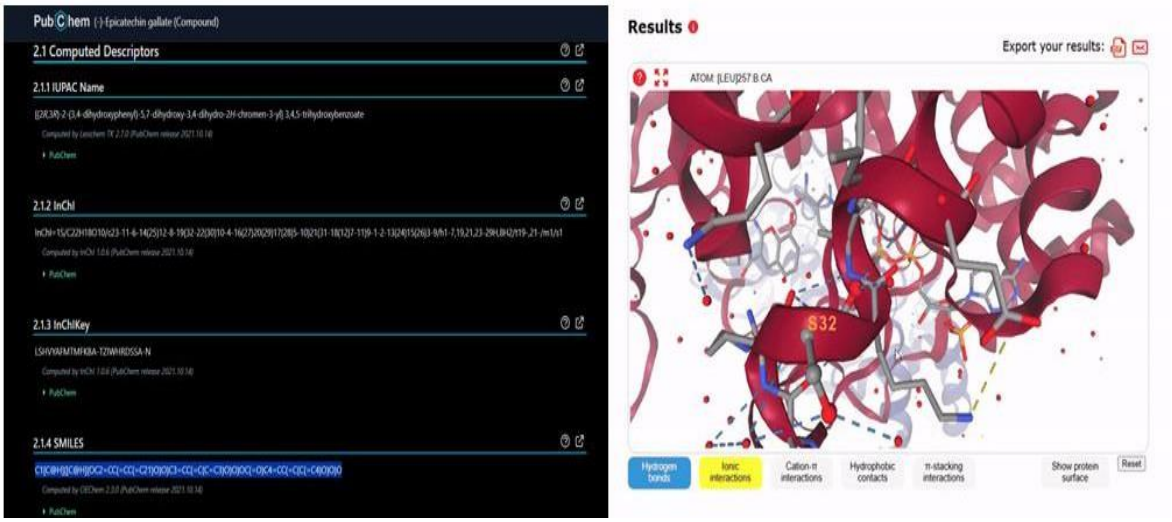


Fig 12: Epicatechin Gallate Smilies identified & docking with 5 Alpha Reductase

MODEL	AFFINITY
1	-5.33
2	-5.1

HYDROGEN BONDS	ASP-GLU
	ASP-LYS
	TYR-ASP
	ALA-ILE
	PHE-LYS
	TYR-HIS

Table 3: Calculated Affinity (Kcal/Mol) And Hydrogen Bonds Identified

EPICATECHIN

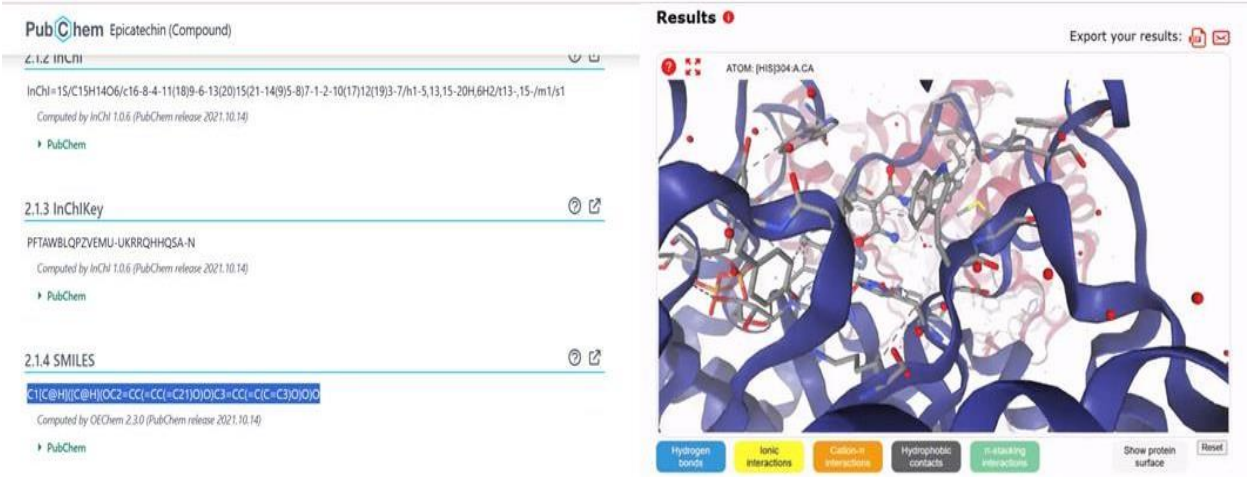


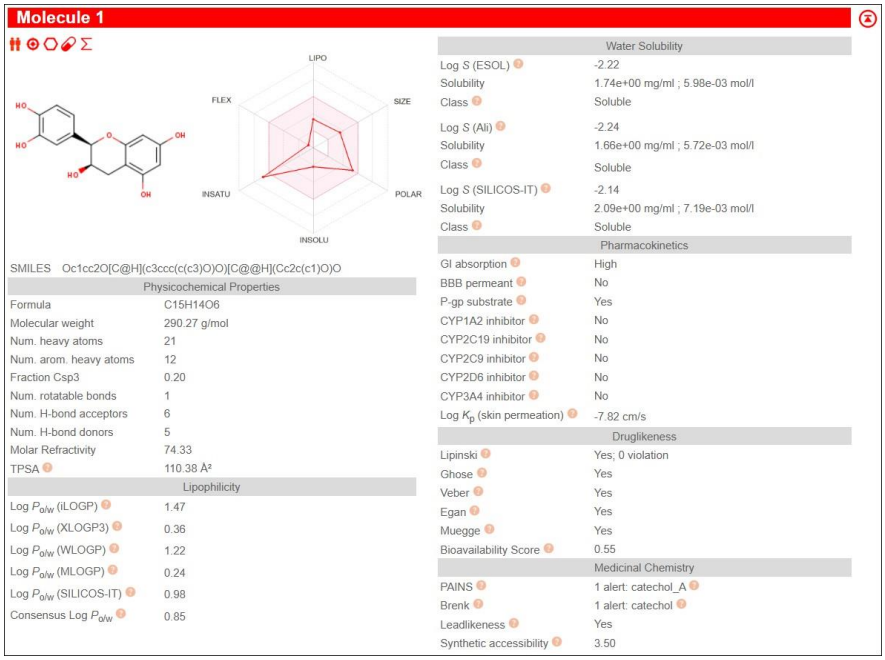
Fig 13: Epicatechin Gallate Smilies identified & docking with 5 Alpha Reductase

MODEL	AFFINITY
1	-3.55
2	-1.22

HYDROGEN BONDS	GLU-ASP
	LYS-GLU
	TYR-ASP
	HIS-ALA
	PHE-LYS
	ALA-HIS

Table 4: Calculated Affinity (Kcal/Mol) and Hydrogen bonds identified

ADME STUDIES FOR DRUG LIKELYHOOD:-



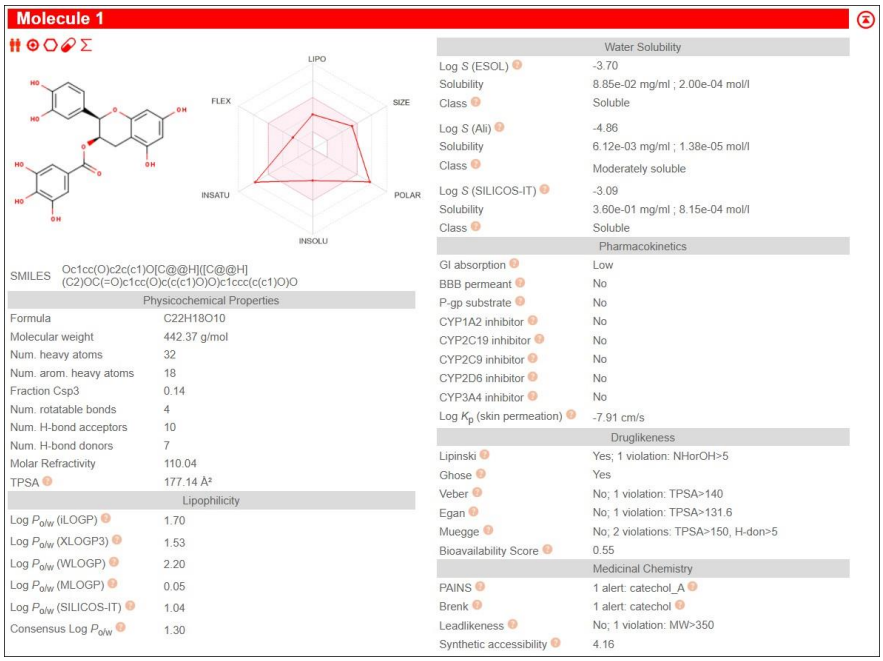
Molecule 1

SMILES Oc1cc2O[C@H](c3ccc(c(c3)O)O)[C@@H](Cc2c(c1)O)O

Physicochemical Properties	
Formula	C15H14O6
Molecular weight	290.27 g/mol
Num. heavy atoms	21
Num. arom. heavy atoms	12
Fraction Csp3	0.20
Num. rotatable bonds	1
Num. H-bond acceptors	6
Num. H-bond donors	5
Molar Refractivity	74.33
TPSA	110.38 Å²
Lipophilicity	
Log $P_{o/w}$ (iLOGP)	1.47
Log $P_{o/w}$ (XLOGP3)	0.36
Log $P_{o/w}$ (WLOGP)	1.22
Log $P_{o/w}$ (MLOGP)	0.24
Log $P_{o/w}$ (SILICOS-IT)	0.98
Consensus Log $P_{o/w}$	0.85

Water Solubility	
Log *S* (ESOL)	-2.22
Solubility	1.74e+00 mg/ml ; 5.98e-03 mol/l
Class	Soluble
Log *S* (Ali)	-2.24
Solubility	1.66e+00 mg/ml ; 5.72e-03 mol/l
Class	Soluble
Log *S* (SILICOS-IT)	-2.14
Solubility	2.09e+00 mg/ml ; 7.19e-03 mol/l
Class	Soluble
Pharmacokinetics	
GI absorption	High
BBB permeant	No
P-gp substrate	Yes
CYP1A2 inhibitor	No
CYP2C19 inhibitor	No
CYP2C9 inhibitor	No
CYP2D6 inhibitor	No
CYP3A4 inhibitor	No
Log K_p (skin permeation)	-7.82 cm/s
Druglikeness	
Lipinski	Yes; 0 violation
Ghose	Yes
Veber	Yes
Egan	Yes
Muegge	Yes
Bioavailability Score	0.55
Medicinal Chemistry	
PAINS	1 alert: catechol_A
Brenk	1 alert: catechol
Leadlikeness	Yes
Synthetic accessibility	3.50

Fig 14: ADME Studies of Catechin



Molecule 1

SMILES Oc1cc(O)c2c(c1)O[C@@H]([C@@H](C2)OC(=O)c1cc(O)c(c(c1)O)O)c1ccc(c(c1)O)O

Physicochemical Properties	
Formula	C22H18O10
Molecular weight	442.37 g/mol
Num. heavy atoms	32
Num. arom. heavy atoms	18
Fraction Csp3	0.14
Num. rotatable bonds	4
Num. H-bond acceptors	10
Num. H-bond donors	7
Molar Refractivity	110.04
TPSA	177.14 Å²
Lipophilicity	
Log $P_{o/w}$ (iLOGP)	1.70
Log $P_{o/w}$ (XLOGP3)	1.53
Log $P_{o/w}$ (WLOGP)	2.20
Log $P_{o/w}$ (MLOGP)	0.05
Log $P_{o/w}$ (SILICOS-IT)	1.04
Consensus Log $P_{o/w}$	1.30

Water Solubility	
Log *S* (ESOL)	-3.70
Solubility	8.85e-02 mg/ml ; 2.00e-04 mol/l
Class	Soluble
Log *S* (Ali)	-4.86
Solubility	6.12e-03 mg/ml ; 1.38e-05 mol/l
Class	Moderately soluble
Log *S* (SILICOS-IT)	-3.09
Solubility	3.60e-01 mg/ml ; 8.15e-04 mol/l
Class	Soluble
Pharmacokinetics	
GI absorption	Low
BBB permeant	No
P-gp substrate	No
CYP1A2 inhibitor	No
CYP2C19 inhibitor	No
CYP2C9 inhibitor	No
CYP2D6 inhibitor	No
CYP3A4 inhibitor	No
Log K_p (skin permeation)	-7.91 cm/s
Druglikeness	
Lipinski	Yes; 1 violation: NHorOH>5
Ghose	Yes
Veber	No; 1 violation: TPSA>140
Egan	No; 1 violation: TPSA>131.6
Muegge	No; 2 violations: TPSA>150, H-don>5
Bioavailability Score	0.55
Medicinal Chemistry	
PAINS	1 alert: catechol_A
Brenk	1 alert: catechol
Leadlikeness	No; 1 violation: MW>350
Synthetic accessibility	4.16

Fig 15: ADME Studies of Epicatechin Gallate

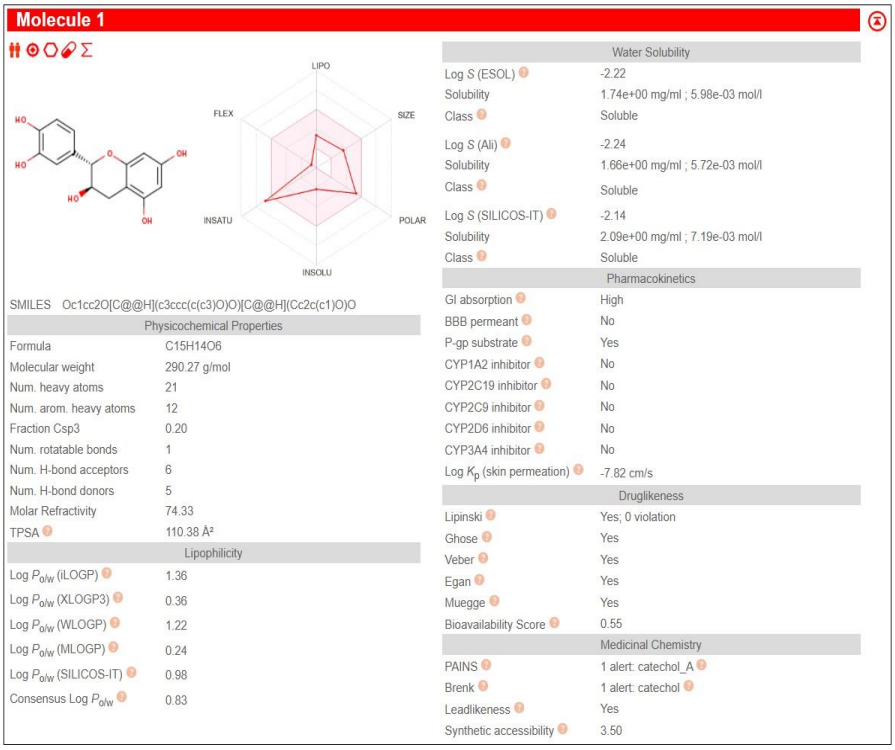
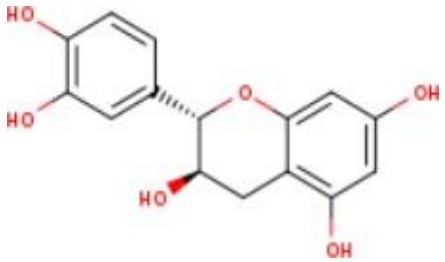
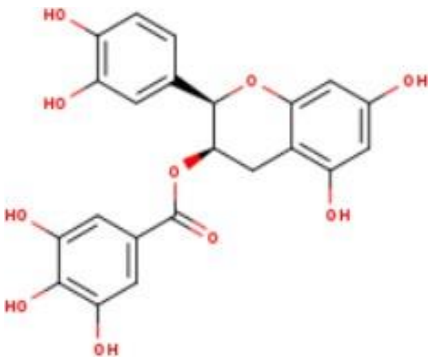
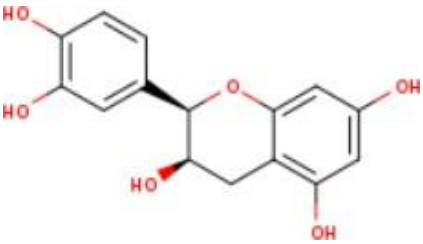


Fig 16: ADME Studies of Epicatechin

ADME RESULTS

Catechin and Epicatechin are strong contenders for drug development because of their favorable ADME profiles, straightforward nature, and ease of synthesis. Although Epicatechin Gallate has some limitations, it presents unique benefits in certain biological situations and deserves further investigation for targeted uses. Strategies to optimize Epicatechin Gallate could aim at improving its solubility and streamlining its synthesis, while Catechin and Epicatechin are prepared for preclinical studies with only minor adjustments.

NAME	STRUCTURE	PROPERTIES
CATACHIN		**Formula:** $C_{15}H_{14}O_6$ **SMILES:**Oc1cc2O[C@@H](c3ccc(c(c3)O)O)[C@@H](Cc2c(c1)O)O **Mol.Wt:** 290.27 g/mol **No. Rotatable bond:** 1 **No. of H-bond acceptors:6 No. of H-bond donors:5Molar Refractivity:** 108.88 **TPSA:** 67.87 A^0 **Log Pa/w(iLOGP):** 1.36 **Log Pa/w(XLOGP3):** 0.36 **Log Pa/w(MLOGP):** 0.24 **Log S(ESOL):** -2.22 **Solubility class:** soluble **Lipinski:** Yes; 0 violation **Bioavailabilty score:** 0.55 **Leadlikeness: Yes Synthetic accessibility:** 3.5
		Formula: $C_{22}H_{18}O_{10}$ **SMILES:**Oc1cc(O)c2c(c1)O[C@@H]([C@@H](C2)OC(=O)c1cc(O)c(c(c1)O)O)c1ccc(c(c1)O)O **Mol.Wt:** 442.37 g/mol **No. Rotatable bond:** 4 **No. of H-bond acceptors:10** **No. of H-bond donors:7** **Molar Refractivity:**

EPICATACHIN GALLATE		110.04 **TPSA:** 177.14 Å **Log Po/w(iLOGP):** 1.70 **Log Po/w(XLOGP3):** 1.53 **Log Po/w(MLOGP):** 0.05 **Log S(ESOL):** -3.70 **Solubility class:** soluble **Lipinski:** Yes; 1 violation NHorOH>5 **Bioavailabilty score:** 0.55 **Leadlikeness:** No; 1 violation: MW>350 **Synthetic accessibility:** 4.16
EPICATACHIN		**Formula:** $C_{15}H_{14}O_6$ **SMILES:**Oc1cc2O[C@@H](c3ccc(c(c3)O)O)[C@@H](Cc2c(c1)O)O **Mol.Wt:** 290.27g/mol **No. Rotatable bond:** 1 **No. of H-bond acceptors:6** **No. of H-bond donors:5** **Molar Refractivity:** 74.33 **TPSA:** 110.38 Å **Log Po/w(iLOGP):** 1.36 **Log Po/w(XLOGP3):** 0.36 **Log Po/w(MLOGP):** 0.24 **Log S(ESOL):** -2.22 **Solubility class:** soluble **Lipinski:** Yes; 0

		violation **Bioavailabilty score:** 0.55 **Leadlikeness:** Yes **Synthetic accessibility:** 3.50

Table 5: ADME Studies Results

COMPARATIVE STUDY:

PROPERTY	**CATACHIN**	**EPICATACHIN**	**EPICATACHIN GALLATE**
Molecular Weight	290.27g/mol	290.27g/mol	442.37g/mol
Rotatable Bonds	1	1	4
H-Bond Acceptors	6	6	10
H-Bond Donors	5	5	7
Solubility	SOLUBLE	SOLUBLE	SOLUBLE
Lipinski Violations	0	1	1
Bioavailability	0.55	0.55	0.55
Lead Likeliness	YES	YES	YES
Synthetic Accessibility	3.50	3.50	4.16

Table 6: ADME Comparative Study

5.5 OIL INFUSION

Oil infusion is a process of extracting the beneficial compounds of herbs, plants, or other natural ingredients by soaking them in carrier oil over a specific period, either with the application of gentle heat or through a cold infusion method. This technique allows the oil to absorb the active constituents, aromas, and nutrients from the infused materials, creating a versatile product for cosmetic, medicinal, or culinary use. Coconut oil is a great oil infusion, which provides a stable medium for extraction and preservation of the bioactive compounds from the areca husk and peduncle. High in lauric acid, this oil has a high capacity to penetrate through the cell membrane and is an excellent carrier of beneficial compounds of natural ingredients. In the oil infusion technique, dried shredded areca husk and peduncle are combined with coconut oil and heated slowly over lower temperatures. The bioactive compounds, like antioxidants, tannins, and alkaloids, diffuse into the oil, imparting therapeutic value.

The infusion is further enhanced by the addition of saw palmetto oil, whose DHT-blocking action is exceptionally strong. DHT is a hormone that causes thinning and loss of hair. Saw palmetto causes hair loss by blocking the enzyme 5-alpha reductase, thus reducing scalp levels of DHT. This powerful infused oil will provide nourishment and strength to hair while working against hair loss. The mixed oil infusion will contribute nutrients that can serve as a natural remedy for promoting hair growth, inhibiting inflammation of the scalp, and improving health for both the hair and skin. The continued use of this oil infusion is a sure support for a comprehensive approach to hair care and long-persistent benefits.

Fig 17: Oil infusion

Fig 18: Collected infused oil

5.6 FORMULATIONS

FORMULA -1: HENNA INDIGO MIXED

SL NO	INGRIDIENT	BOTANICAL NAME	QUANTITY
1	HENNA POWDER	*Lowsoniainermis*	20g
2	INDIGO POWDER	*Indigoferatinctoria*	15g
3	BRINGRAJ POWDER	*Ecliptaprostrata*	1g
4	AMLA POWDER	*Phyllanthusemblicia*	1g
5	INFUSED OIL		1ml
6	ALOE VERA GEL	*Aloe barbadensis*	2ml

Table 7: Henna Indigo Mixed Formulation

1. Preparation of Powders

Make sure all the powders (Henna, Indigo, Bhringraj, Amla) are finely ground and free from impurities. If needed, use a sieve to eliminate any coarse particles.

2. Mixing the Powders

In a mixing bowl, combine the measured amounts of Henna Powder (20 g), Indigo Powder (15 g), Bhringraj Powder (10 g), and Amla Powder (12 g).

Stir well to ensure that all the powders are evenly distributed.

3. Incorporation of Infused Oil

Gradually add the infused oil (the exact quantity needs to be specified). Mix it into the powders until you achieve a uniform blend.

The infused oil serves as a medium to improve the texture and effectiveness of the final product.

4. Adding Aloevera Gel

Incorporate 2 mL of Aloe Vera Gel (Aloe barbadensis) into the mixture. Blend thoroughly to ensure the gel is well integrated. Aloevera Gel adds hydration and helps create a smooth consistency.

5. Hydration

Slowly introduce water or another hydrating agent to the mixture until you reach a paste-like consistency that is suitable for application.

Allow the mixture to sit for at least 30 minutes so the powders can absorb the moisture and release their beneficial properties

FORMULA- 2: HENNA SOACKED IN WATER+INDIGO

SL NO	INGRIDIENT	BOTANICAL NAME	QUANTITY
1	HENNA POWDER	*Lowsoniainermis*	20g
2	INDIGO POWDER	*Indigoferatinctoria*	15g
3	BRINGRAJ POWDER	*Ecliptaprostrata*	1g
4	AMLA POWDER	*Phyllanthusemblicia*	1g
5	INFUSED OIL		1ml
6	ALOE VERA GEL	*Aloe barbadensis*	2ml

Table 8: Henna Socked in Water + Indigo Formulation

Step 1: Preparation of formulated Henna Paste

Mixing Ingredients:

Combine 20 g of Henna Powder (Lawsoniainermis) with:

1 g of Bhringraj Powder (Ecliptaprostrata)

1 g of Amla Powder (Phyllanthusemblica)

Add 1 mL of infused oil to the powder mixture

Add 2ml of Aloevera gel then gently stir well so that all ingredients are well mixed together and add water to maintain the consistency of the paste.

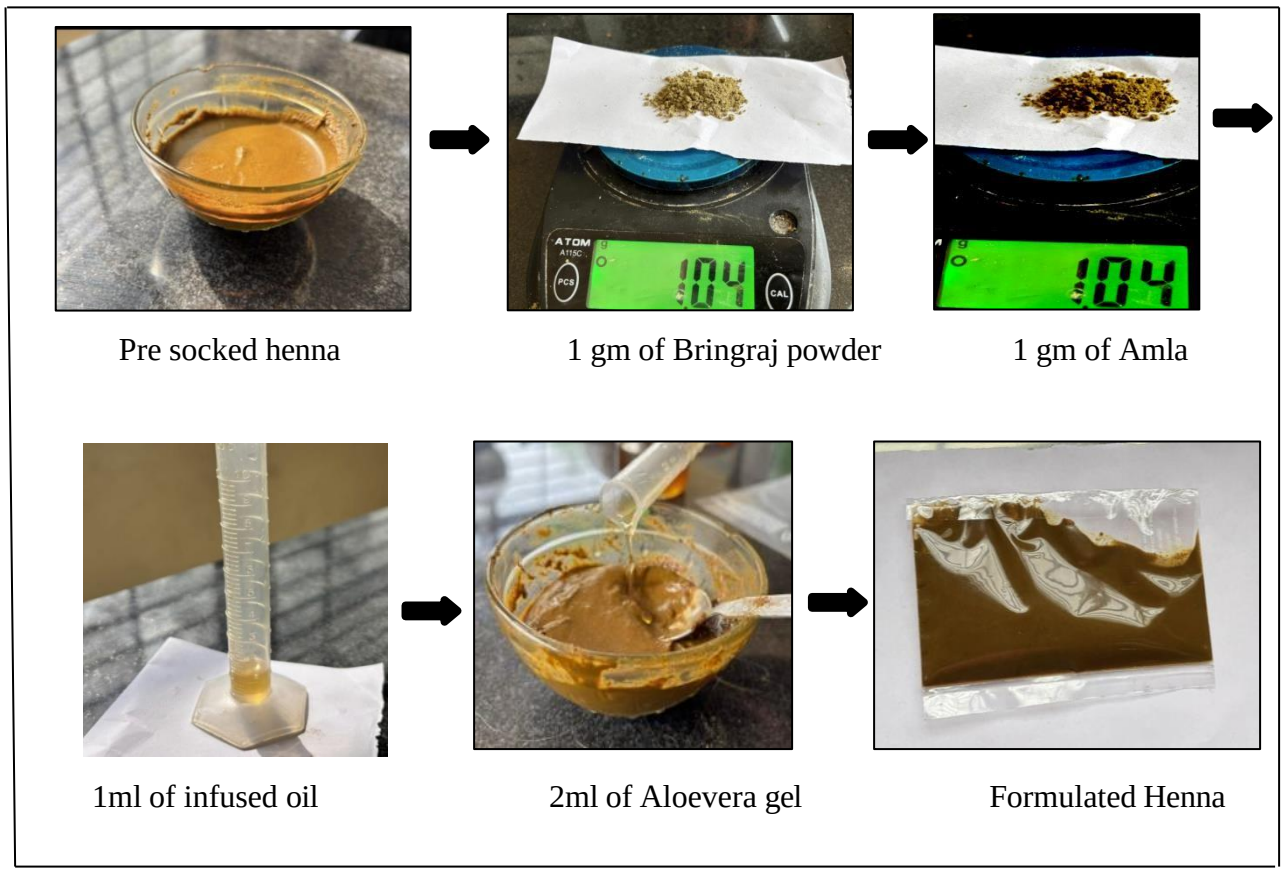


Fig 19: Formulated henna paste

Fig 20: Henna + indigo hair dye

STEPS TO APPLY THE HAIR DYE

Application of Henna Paste

Apply the henna paste evenly to clean, dry hair, making sure to cover from root to tip. Leave the paste on for 2-3 hours to ensure proper dye absorption. Rinse thoroughly with lukewarm water (no shampoo). Allow your hair to air dry.

Preparation of Indigo Paste

Mix 15 g of Indigo Powder (*Indigoferatinctoria*) with warm water to create a thick paste. Let the paste sit for 15-20 minutes to activate.

Application of Indigo Paste

Apply the indigo paste to freshly washed hair after the henna treatment, ensuring even coverage. Leave the paste on for 1-2 hours, depending on how intense you want the color to be. Allow your hair to air dry naturally during the application.

Final Rinse

Rinse your hair thoroughly with water to remove the indigo paste. Avoid shampooing for at least 24-48 hours to let the color set properly.

FORMULA- 3: HENNA SOCKED IN TEA DECOCTION WITH INDIGO PASTE

SL NO	INGRIDIENT	BOTANICAL NAME	QUANTITY
1	HENNA POWDER	*Lowsonia inermis*	25g
2	INDIGO POWDER	*Indigofera tinctoria*	15g
3	BRINGRAJ POWDER	*Eclipta prostrata*	1g
4	AMLA POWDER	*Phyllanthus emblicia*	1g
5	INFUSED OIL		1ml
6	ALOE VERA GEL	*Aloe barbadensis*	2ml

Table9: Henna Socked In Tea Decoction + Indigo Formulation

Preparation of Henna Paste

Prepare the Tea Extract:

Boil water and steep 1-2 teaspoons of tea leaves for 10-15 minutes to create a strong tea solution. Strain it and let it cool to a lukewarm temperature.

Mix Henna with Tea Extract:

Combine 25 g of Henna Powder (*Lawsoniainermis*) with the cooled tea extract. Gradually add the liquid to create a smooth, thick paste. Add 1g of Bringraj powder,1g of Amla powder and Infused oil into the mixture. Next ,1-2 drops of Aloe Vera Gel (2 mL) for extra hydration and a smoother application. Let the paste sit for 4-6 hours (or overnight) to allow the dye to release.

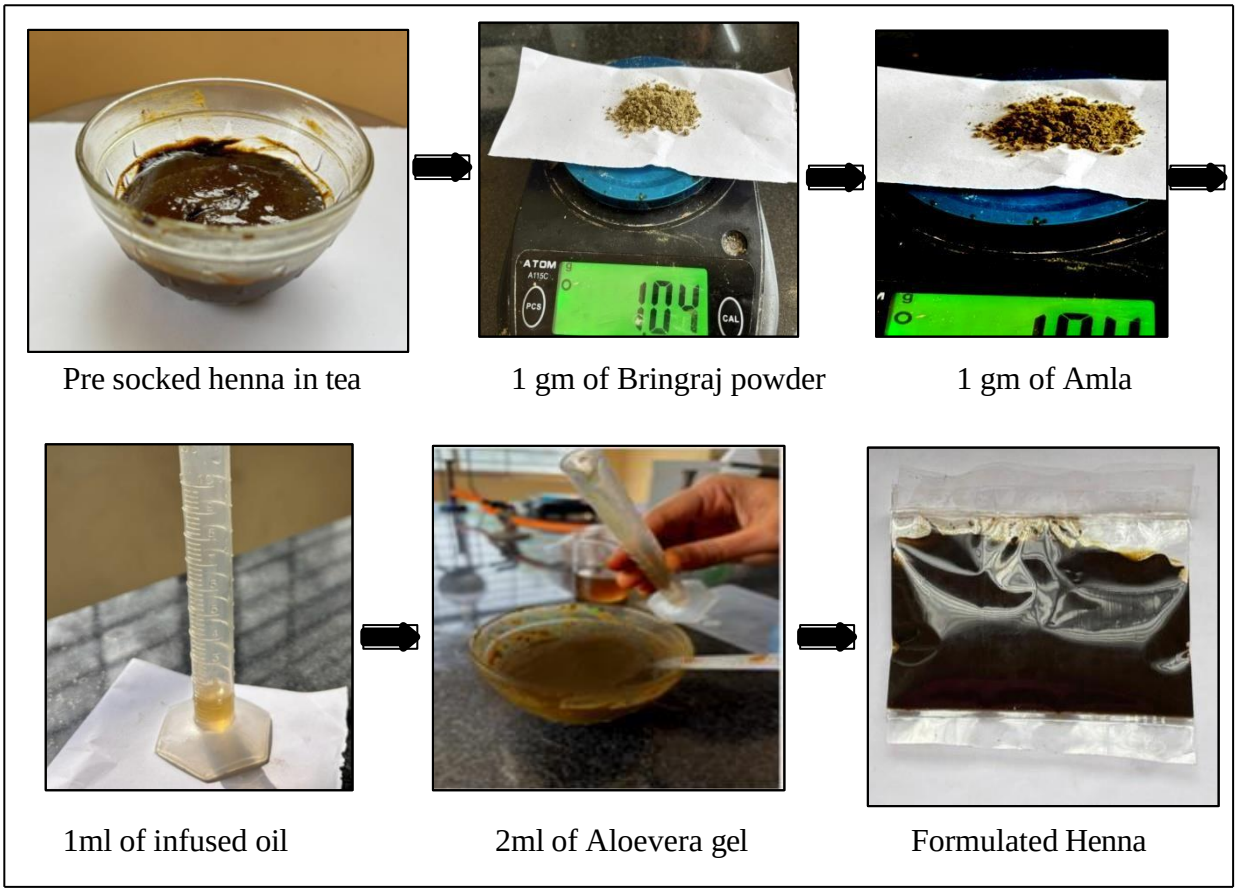


Fig 21: Formulated henna paste in Tea decoction

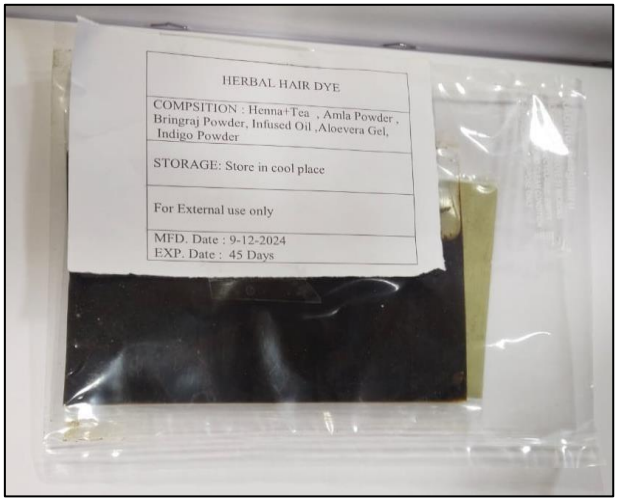


Fig 22: Henna in tea + indigo hair dye

STEPS TO APPLY THE HAIR DYE

Application of Henna Paste

Apply the henna paste evenly to clean, dry hair, making sure to cover from root to tip. Leave the paste on for 2-3 hours to ensure proper dye absorption. Rinse thoroughly with lukewarm water (no shampoo). Allow your hair to air dry.

Preparation of Indigo Paste

Mix 15 g of Indigo Powder (*Indigoferatinctoria*) with warm water to create a thick paste. Let the paste sit for 15-20 minutes to activate.

Application of Indigo Paste

Apply the indigo paste to freshly washed hair after the henna treatment, ensuring even coverage. Leave the paste on for 1-2 hours, depending on how intense you want the color to be. Allow your hair to air dry naturally during the application.

Final Rinse

Rinse your hair thoroughly with water to remove the indigo paste. Avoid shampooing for at least 24-48 hours to let the color set properly.

FORMULA- 4: SEMISYNTHETIC FAST ACTING FORMULA WITH 5% PPD

SL NO	INGRIDIENT	BOTANICAL NAME	QUANTITY
1	HENNA POWDER	*Lowsoniainermis*	25g
2	INDIGO POWDER	*Indigoferatinctoria*	15g
3	BRINGRAJ POWDER	*Ecliptaprostrata*	1g
4	AMLA POWDER	*Phyllanthusemblicia*	1g
5	INFUSED OIL		1ml
6	ALOE VERA GEL	*Aloe barbadensis*	2ml
7	PPD	-	3g+2ml H2o2

Table10: Semi Synthetic Fast Acting Formula

Preparation of synthetic dye: Combine 20g of Henna Powder (*Lawsoniainermis*) with 1g of Bhringraj Powder (*Ecliptaprostrata*) 1g of Amla Powder (*Phyllanthusemblica*) Add 1 ml of infused oil to the powder mixture along with indigo powder . Add 2 ml PPD and 3 ml of H_2O_2 the solution.

Formulated henna+ indigo mix 3g of PPD 2ml of hydrogen peroxide

Fig 23: Synthetic hair dye formulation

Fig 24: Synthetic hair dye

STEPS TO APPLY SYNTHETIC HAIR DYE

To apply synthetic hair dye, first, perform a patch test 24–48 hours prior to ensure no allergic reaction. Apply the synthetic hair dye and leave it for 30 minutes then rinse the hair with the water

EVALUVATION

ORANOLEPTIC EVALUVATION

SL NO	PARAMETER	HENNA INDIGO MIX	HENNA MIX SOACKED IN WATER	HENNA MIX SOACKED IN TEA EXTRACT	SEMI-SYNTHETIC
1	Colour	Black	Dark Green	Brownish Green	Black
2	Odour	Characteristic	Characteristic	Tea Odour	No
3	Texture	Semi Solid	Semi Solid	Semi Solid	Semi Solid
4	Appearance	Paste	Paste	Paste	Paste

Table11:Organoleptic Evaluation

PATCH TEST

SL NO	PARAMETER	HENNA INDIGO MIX	HENNA MIX SOACKED IN WATER	HENNA MIX SOACKED IN TEA EXTRACT	SEMI-SYNTHETIC
1	Swelling	Negative	Negative	Negative	Negative
2	Redness	Negative	Negative	Negative	Negative
3	Irritation	Negative	Negative	Negative	Negative

Table12: Patch Test Results

STABILITY TEST

Sl No	Parameter	Henna Indigo Mix		Henna Socked In Water		Henna socked In Tea Extract		Semi-Synthetic	
		20°C	36°C	20°C	36°C	20°C	36°C	20°C	36°C
1	Colour	No Change	No Change	No Change	No Change	No Change	No Change	No Change	No Change
2	Odor	No Change	No Change	No Change	No Change	No Change	No Change	No Change	No Change
3	PH	-	-	5.04	5.04	6.3	6.4	5.6	5.7
4	Texture	Smooth	Smooth	Smooth	Smooth	Smooth	Smooth	Smooth	Smooth

Table 13: Stability Test Results

COMPARISON OF COLOURING EFFECT OF HERBAL HAIR DYE vs MARKETED BRAND ON HUMAN HAIR

Name Of Hair Dye	Colour Effect	Duration Of Exposure	Physical Appearance
Henna Indigo Mix	**2/5**	3 Hour UntillAirdried	Smooth No Damage
Henna Mix In Water	**5/5**	3 Hour UntillAirdried	Smooth No Damage
Henna Mix In Tea Extract	**5/5**	3 Hour UntillAirdried	Smooth No Damage
Semi synthetic	**5/5**	30 Minutes	No Damage But Rough Hair Cortex Observed
Synthetic	**5/5**	15 Minutes	Cortex Damaged

Table14: Comparison Results

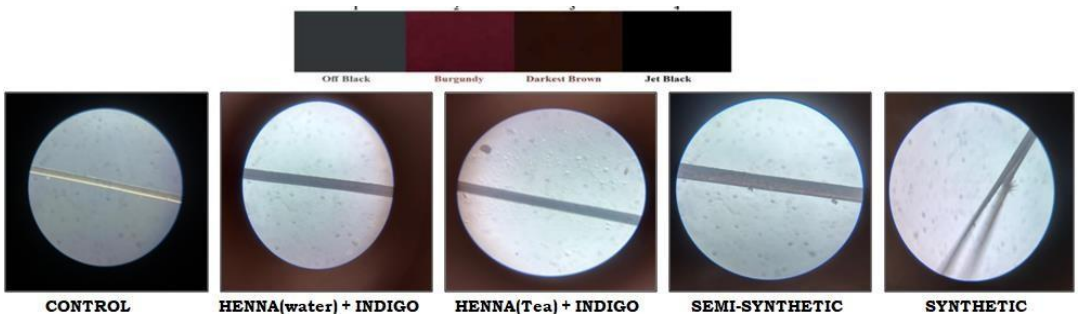


Fig 25: Semi Synthetic Fast Acting Formula

Fig 26: Test Results of Hair Dying Effects on Real Human Hair

Fig27: Final Packed Formulation of Hair Dye

DISCUSSIONS

ORGANOLEPTIC EVALUVATION

The organoleptic evaluation of four hair dye formulations revealed differences in color, odor, and appearance. Henna-Indigo and semi-synthetic dyes showed black color, while water-soaked henna was dark green, and tea-soaked henna was brownish green. All had semi-solid texture with paste-like appearance, while only tea extract imparted a distinctive tea odor.

PATCH TEST

The patch test results for all four hair dye formulations (Henna-Indigo mix, water-soaked henna mix, tea-soaked henna mix, and semi-synthetic dye) indicated no adverse reactions. Parameters such as swelling, redness, and irritation were negative for all formulations, demonstrating their skin compatibility and safety for topical application.

STABILITY TEST

The stability test for four hair dye formulations (Henna-Indigo mix, water-soaked henna, tea-soaked henna, and semi-synthetic dye) showed no changes in color, odor, or texture at both 20°C and 36°C. PH values ranged from 5.04 to 6.4, with consistent smooth texture across all formulations, indicating stability under varying conditions.

COMPARISON WITH SYNTHETIC HAIR DYES IN MARKET

The herbal hair dye requires a longer application time of 3 hours and ensures smooth application with no damage to the hair cortex. In contrast, semi-synthetic and synthetic dyes are fast-acting formulas. While the semi-synthetic dye causes slight roughness to the hair cortex, the synthetic dye results in significant cortex damage after 25 washes.

APPLICATIONS

Applications of a Nourishing Hair Dye with Natural Color and Hair Care Benefits

Natural Coloring and Shine Enhancement

Henna + Indigo + Tea Extract: Offers a vibrant, natural hair color with long-lasting results. Henna conditions and strengthens the hair, while Indigo adds a darker tone. Tea extract enriches the color and enhances the natural shine, leaving hair lustrous and soft.

Scalp Nourishment and Hydration

Aloe Vera + Bhringraj: Deeply hydrates and nourishes the scalp, reducing dryness, itchiness, and dandruff. Bhringraj improves blood circulation to hair follicles, creating a healthy environment for hair growth.

5-Alpha Reductase Inhibition for Hair Loss Prevention

Areca (Catechin, Epicatechin, Epicatechin Gallate): Blocks the activity of 5-alpha reductase, reducing DHT production. This helps prevent hair follicle miniaturization and supports thicker, healthier hair growth.

DHT Blocking for Stronger Hair

Saw Palmetto: Acts as a natural DHT blocker, preventing hair loss associated with androgenic alopecia. It strengthens the hair from the roots, reducing breakage and thinning.

UV Protection and Antioxidant Support

Amla + Areca + Bhringraj: Provides antioxidant protection against UV rays and environmental damage. These ingredients help maintain hair integrity, prevent premature graying, and improve overall hair resilience

FUTURE SCOPE

Future Scope of Work for the Nourishing Herbal Hair Dye

Toxicology Studies via Animal Trials

Conduct comprehensive toxicology assessments using animal models to ensure the safety of the formulation. Focus on potential skin irritations, allergic reactions, and long-term effects, adhering to ethical research standards.

Human Trials with Volunteers

Perform clinical trials involving human volunteers to evaluate the efficacy, safety, and user satisfaction. Assess parameters such as hair coloring performance, reduction in hair loss, scalp health improvements, and any adverse reactions.

Standardization and Quality Control

Develop standardized protocols for ingredient sourcing, preparation, and final product quality testing. This ensures consistent performance and safety across all batches of the product.

Patenting the Formulation

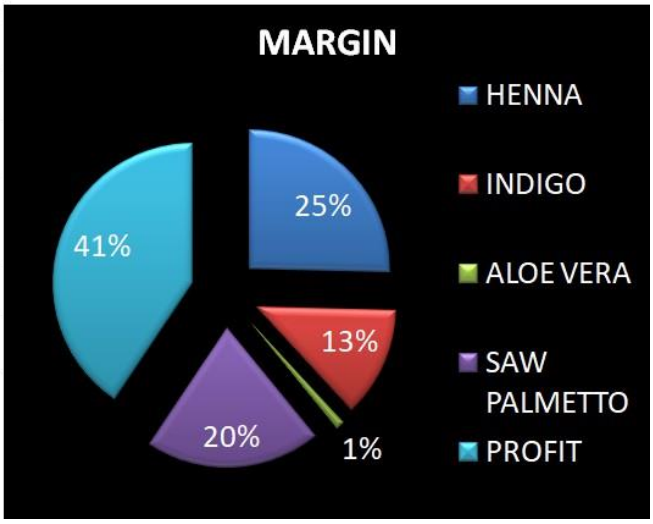
Secure intellectual property rights for the innovative blend of natural ingredients used for coloring, nourishing, and therapeutic benefits. This step protects the product from duplication and establishes its uniqueness in the market.

Marketing and Commercialization

Create a marketing strategy emphasizing the product’s natural, multipurpose benefits, including chemical-free hair dyeing, hair loss prevention, and scalp nourishment. Target eco-conscious consumers, salons, and healthcare markets while expanding globally through online and offline channels.

COST FOR 50g PACKET:

HENNA	₹10
INDIGO	₹5
ALOE VERA GEL	₹0.4
BRINGRAJ POWDER	₹0.5
ARECA HUSK	₹0
SAW PALMETTO	₹8



IF COMPARITIVE MARKET PRICE IS KEPT AT ₹40.

Fig 28: Market Analysis of The Product

6. CONCLUSION

In conclusion, the herbal hair dye combines the best of nature's offerings to provide a comprehensive solution for hair care and coloring. Infused with ingredients like **Henna, Indigo, and Tea Extract** for vibrant and long-lasting natural color, and enriched with **Aloe Vera** and **Bhringraj** for scalp nourishment and hydration, this formulation not only beautifies but also promotes hair health. The inclusion of **Saw Palmetto** and **Areca Catechins** ensures advanced hair loss prevention by blocking DHT and inhibiting 5-alpha reductase. With **Amla, Areca, and Bhringraj** offering UV protection and antioxidant benefits, this product stands out as a multi-functional and eco-friendly alternative to synthetic dyes. Future work in toxicology, human trials, and patenting will pave the way for its commercialization, ensuring its safety, efficacy, and global impact

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